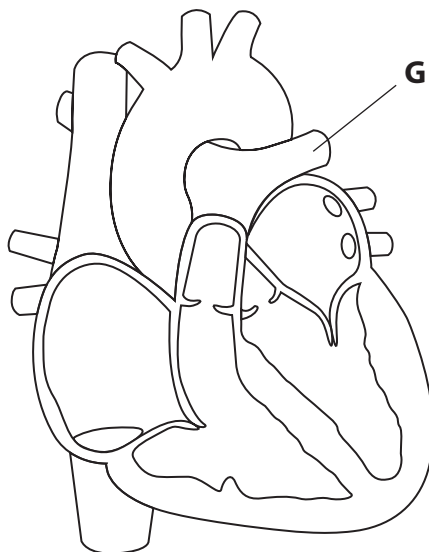


2 The cardiac cycle describes the events that take place in the heart during one complete heartbeat.

(a) The diagram shows a heart in one of the stages, stage **F**, of the cardiac cycle.



(i) What is the name of the blood vessel labelled **G**?

(1)

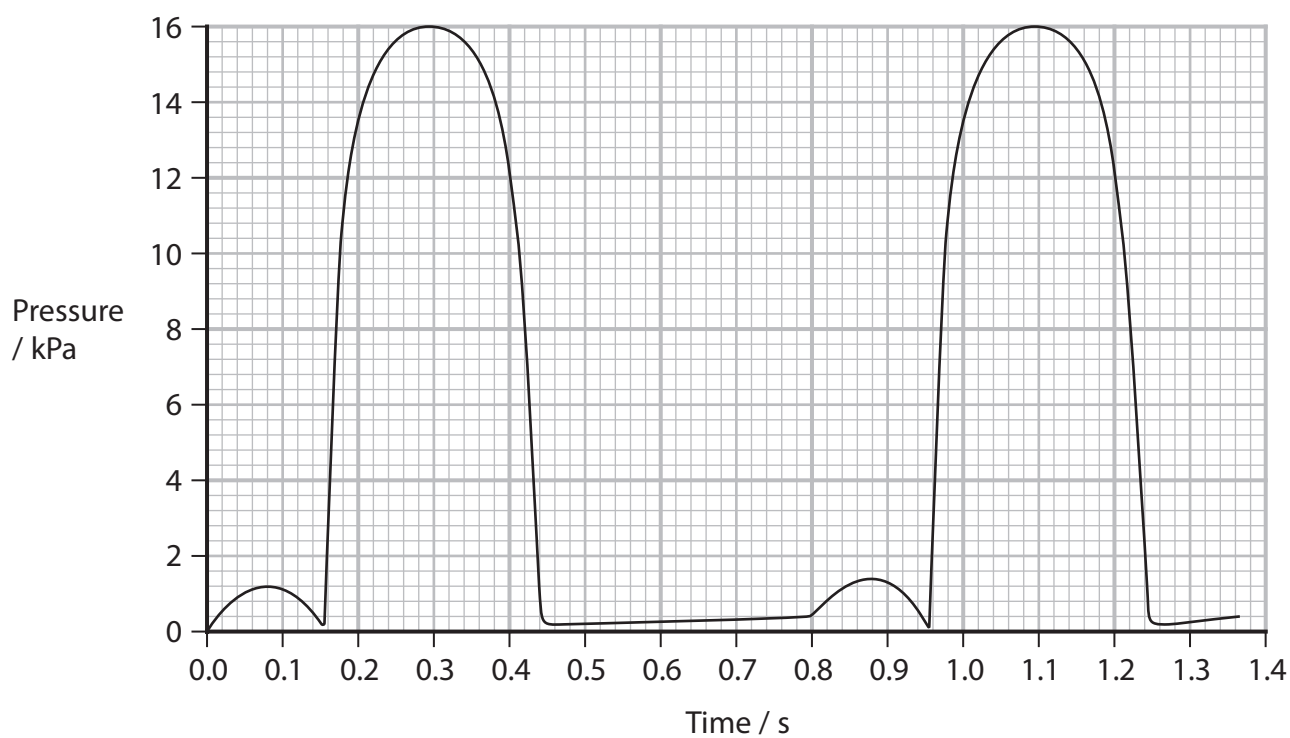
- ☐ **A** aorta
- ☐ **B** pulmonary artery
- ☐ **C** pulmonary vein
- ☐ **D** vena cava

(ii) Which row of the table identifies the stages before and after stage **F**?

(1)

|                                   | Stage before stage F | Stage after stage F |
|-----------------------------------|----------------------|---------------------|
| <input type="checkbox"/> <b>A</b> | atrial systole       | cardiac diastole    |
| <input type="checkbox"/> <b>B</b> | atrial systole       | ventricular systole |
| <input type="checkbox"/> <b>C</b> | ventricular systole  | atrial systole      |
| <input type="checkbox"/> <b>D</b> | ventricular systole  | cardiac diastole    |

(b) The graph shows pressure changes in the left ventricle of a person.



(i) Calculate the heart rate of this person.

Express your answer to 3 significant figures.

(2)

Answer ..... beats  $\text{min}^{-1}$



- (ii) Another line could be drawn on this graph to show the pressure changes in the right ventricle.

Describe the shape and position of this line.

Give reasons for your answer.

(3)

(Total for Question 2 = 7 marks)

